

Faster Is Not Always Greener

The Costs of LLM-Based Code Optimization

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What We Asked Is it worth it to optimize the energy efficiency of code using LLMs?

Why We Asked LLMs are more and more used to optimize the energy efficiency of software **but** LLMs require a lot of energy to run. Therefore, the cost of using LLMs may outweigh their benefits.

What We Found Depending on the configuration, **10% to 58%** of the optimizations are either **incorrect or slower**. The best configurations require **the most energy** to run. **80%** of optimizations require at least **10,000 executions** of the program to be profitable.

For an optimization to be worth it, we should be careful about where and how we apply it.

Break-Even Point (BEP)

The BEP is the number of runs needed before an optimization is **energetically profitable**.

$$\frac{\text{Energy cost of optimization}}{\text{Energy saved per run}}$$

Fig A. Energy cost of optimizing

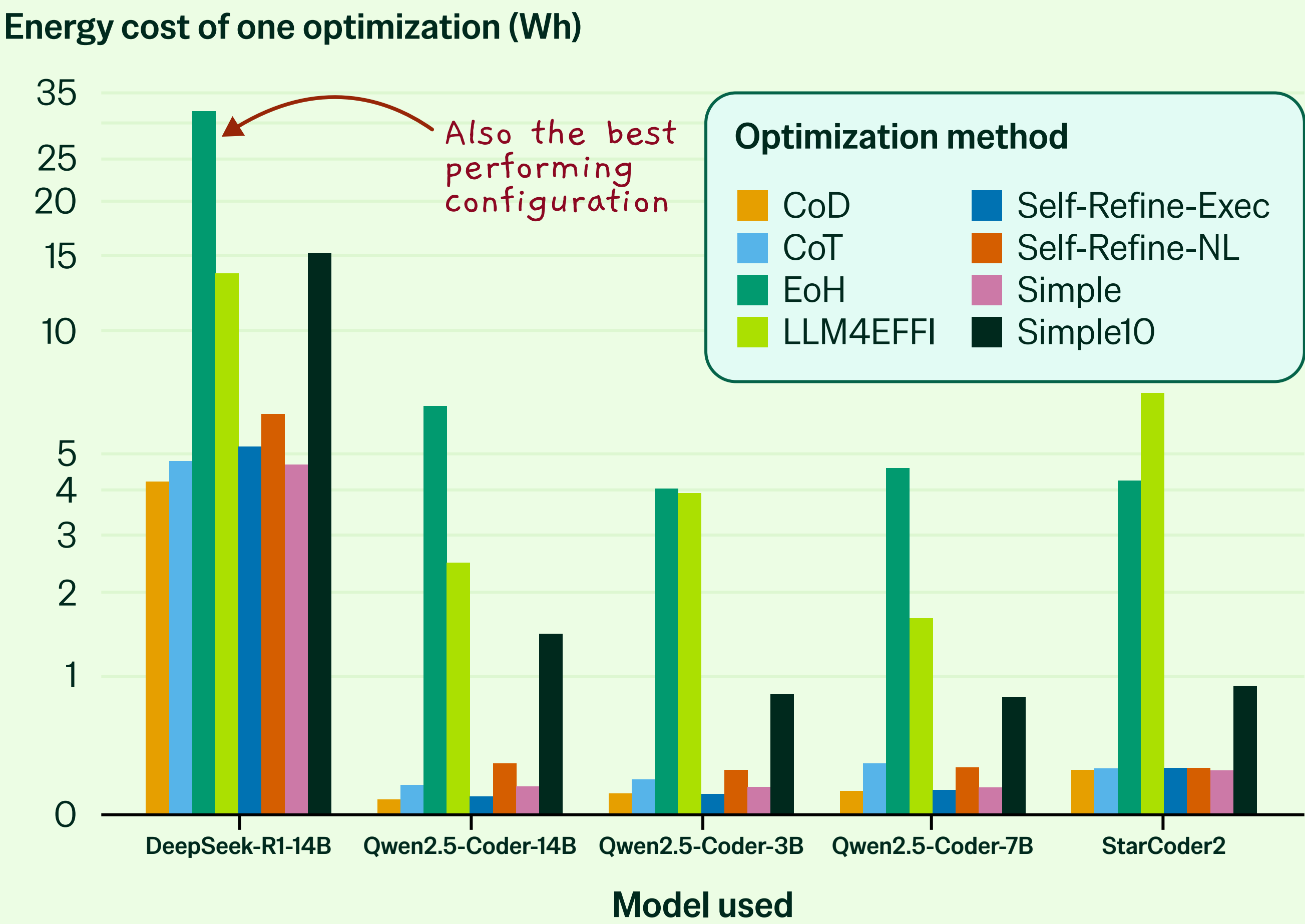
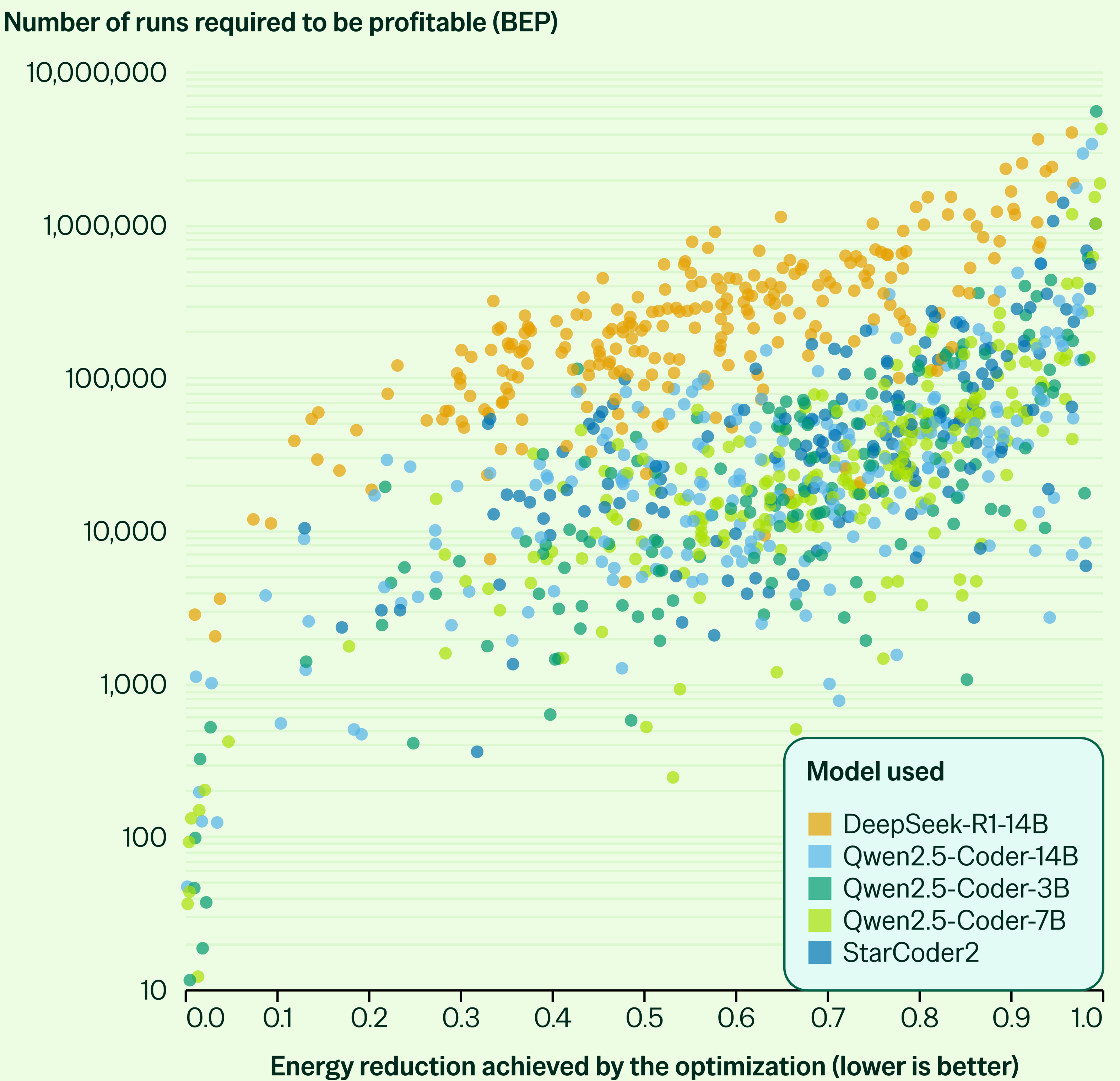


Fig B. Profitability of the optimizations



How We Did It

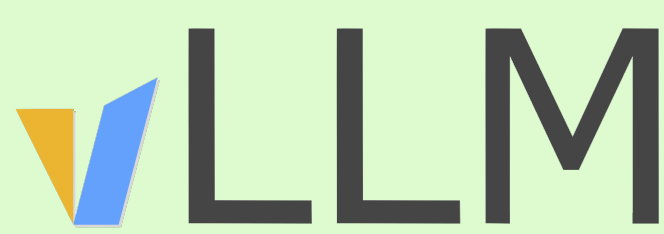
Optimizers & LLMs

8 optimization methods
+
4 standard & 1 reasoning LLMs

Dataset

118 performance-exercising tasks from **EvalPerf**.

Execution



Measures

perf
+
nvidia-smi